Workshop 2 – Day 1

The Merged Model Framework

Recap and practical use



UNIVERSITY OF COPENHAGEN

Jonathan Wenzel Pedersen

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The second workshop

- **Day 1: Recap and practical use**
  Run the model, change an assumption and interpret the results.
- **Day 2: Disaggregation**
  Split a variable into components and reconnect them to the model.
- **Day 3: New relationships**
  Introduce an equation and examine how it changes the model.

Today we return to the model and the code from the first workshop.

Today's programme

1. **Getting started**
   Set up Julia and run the model.

2. **Recap of the Merged Model**
   Revisit the economic relationships, assumptions and model closure.

3. **Working with the model in Julia**
   Find the inputs, assumptions and equations.

4. **Scenarios and interpretation**
   Change an assumption and explain the results.

5. **Feedback and applications**

Getting the model running

1. Clone the repository from GitHub.
2. In VS Code, select **File** > **Open Folder** and open the `Model` folder inside the cloned repository.
3. Press `Ctrl + Shift + P`, search for `Julia: Start REPL`, and press Enter.

The Julia REPL

A terminal should open with the prompt `julia>`. This is where we enter Julia commands.

Activating the project and running the model

1. At the `julia>` prompt, type `]` to enter package mode.
2. Type `activate .` and press Enter. The dot selects the current `Model` folder.
3. Type `instantiate` and press Enter. Wait until installation and precompilation finish.
4. Press Backspace at the empty package prompt to return to `julia>`.
5. Open `run.jl` and click inside the code editor. Press `Ctrl + A` to select all the code, then `Ctrl + Enter` to execute it in the Julia REPL.
6. When execution finishes, open `Merged_model/output/simulation_report.html`.

While I help with setup

If your model is running, explore the project and discuss the following questions with your neighbour.

1. Find the **base year** and **projection horizon**. Which years does the model project?
2. Find the **GDP growth assumptions**. Are they the same for agriculture, industry and services?
3. Find the **equation files**. Can you locate the equation that adds sectoral GDP together to obtain total GDP?
4. Open the **simulation report**. Find GDP, public investment and private investment. How do they develop over the projection?

Keep the baseline unchanged for now. Write down any questions you would like us to revisit.

Model recap

From a growth target to a scenario

A growth target needs a consistent macroeconomic scenario.

- Growth implies requirements for investment and imports.
- Government expenditure must fit with revenue and financing.
- Money, domestic credit and foreign financing must be consistent.

RMSM and Financial Programming: a reminder

RMSM

Growth and export paths imply investment, imports and a real-resource balance.

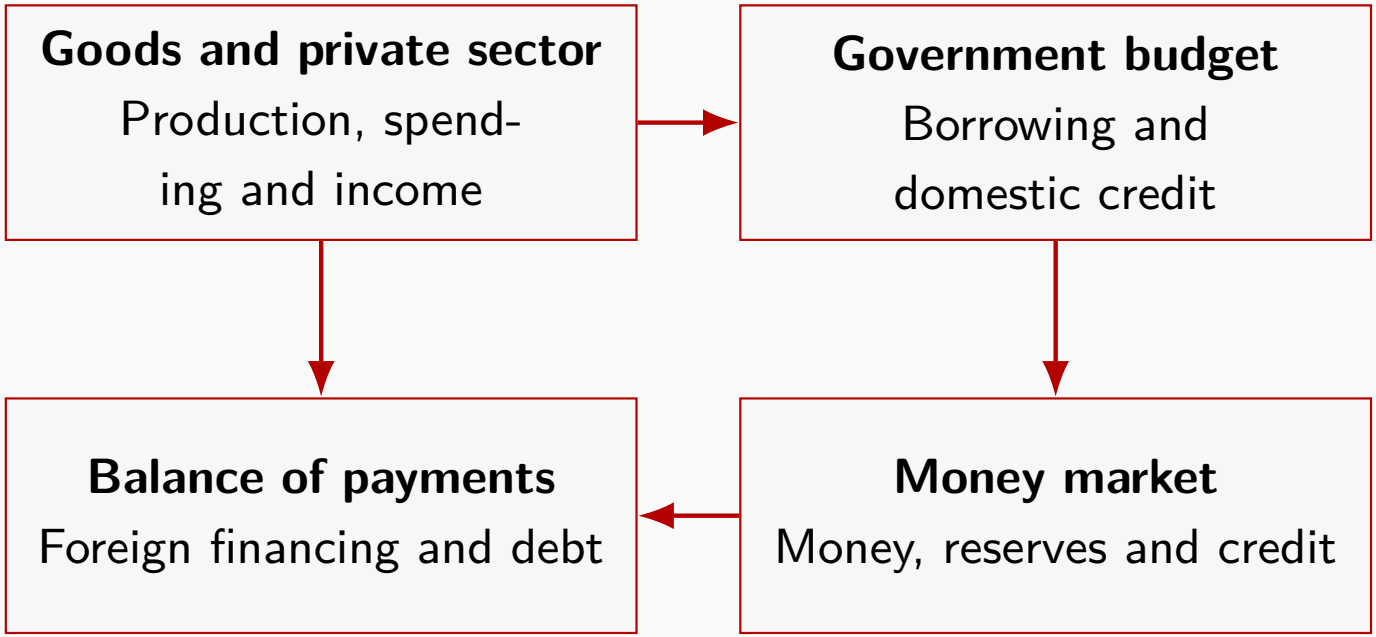
Financial Programming

Government finance, money, credit and external finance must fit together.

The Merged Model

Connects the real-side requirements with the fiscal, monetary and external accounts.

Four connected blocks



Production and exports

Code equation(s): 1–4

$$GDPS_{s,t} = (1 + \gamma_{s,t}) GDPS_{s,t-1}, \qquad GDP_t = \sum_s GDPS_{s,t}$$

$$XS_{s,t} = (1 + xgr_{s,t}) XS_{s,t-1}, \qquad X_t = \sum_s XS_{s,t}$$

- We choose sectoral growth assumptions γ and xgr.
- Sectoral paths add up to aggregate GDP and exports.

Remember GDP growth is an input not a output of this model

Investment and imports

Code equation(s): 5–6

$$IV_t = k_{0,t} GDP_{t-1} + k_{1,t}(GDP_t - GDP_{t-1})$$

Investment depends on the previous GDP level and the change in GDP.

$$\log M_t = m_{0,t} + m_{1,t} \log GDP_t + m_{2,t} \log\left(\frac{E_t MPI_t}{PD_t}\right)$$

Imports depend on GDP and the price of imports relative to domestic goods.

In the code, `IV` denotes total investment and `M` denotes imports.

Consumption and the goods balance

Code equation(s): 7, 9, 11

$$P_0 GDP_t = PD_0(C_t + IV_t) + E_0(XPI_0 X_t - MPI_0 M_t)$$

The constant-price goods balance determines total consumption.

$$PD_t CP_t = (1 - b_t) GDY_t$$

Private disposable income determines private consumption.

$$\boxed{CG_t = C_t - CP_t}$$

Focal variable: government consumption

Government consumption adjusts to the remaining resources.

Does the scenario leave a plausible path for government consumption?

Financing the government

Code equation(s): 15–16

$$NFDG_t = g_t XPI_t X_t$$

Government foreign debt follows the export-linked rule. Non-bank domestic debt $NDDG_t$ follows an assumed path.

$$\boxed{\Delta DCG_t = BRG_t - E_t \Delta NFDG_t - \Delta NDDG_t}$$

Government domestic credit covers the borrowing requirement left after the other sources of financing.

Focal variable: government domestic credit

Is the required expansion in government domestic credit plausible?

$\Delta DCG_t = DCG_t - DCG_{t-1}$ is the change in the credit stock.

Money and domestic credit

Code equation(s): 17–19, 21–22

$$GDPN_t = P_t GDP_t, \qquad MD_t = \frac{GDPN_t}{v_t}, \qquad MS_t = MD_t$$

$$\Delta DC_t = \Delta MS_t - \Delta(E_t R_t)$$

Money demand and reserve holdings determine the required change in total domestic credit.

$$\boxed{DCP_t = DC_t - DCG_t}$$

Focal variable: private domestic credit

Private domestic credit is what remains after government credit.

Does the scenario leave a plausible credit path for the private sector?

External transactions: the current account

Code equation(s): 23–25

$$RESBAL_t = XPI_t X_t - MPI_t M_t$$

$RESBAL$: trade balance; X, M: export and import volumes; XPI, MPI: export and import price indices.

$$NETFSY_t = NFP_t - INFG_t - INFP_t$$

NFP: net factor income excluding foreign interest; $INFG$, $INFP$: government and private foreign interest payments; $NETFSY$: net factor income after these payments.

$$CURBAL_t = RESBAL_t + NETFSY_t + NTRG_t + NTRP_t$$

$CURBAL$: current-account balance; $NTRG$, $NTRP$: net transfers from abroad to government and the private sector.

External financing

Code equation(s): 26

$$\boxed{\Delta NFDP_t = \Delta R_t - CURBAL_t - \Delta NFDG_t - FDI_t}$$

- $NFDP$: private foreign debt.
- R: foreign reserves.
- $CURBAL$: current-account balance.
- $NFDG$: government foreign debt.
- FDI: foreign direct investment inflows.

Focal variable: private foreign debt

Private foreign borrowing covers the remaining financing need. Is the implied debt path plausible?

What do the focal variables tell us?

The model calculates the adjustments required by our growth and policy assumptions.

Government		**Private sector**	
CG_t	DCG_t	DCP_t	$NFDP_t$
Consumption and domestic credit		Domestic credit and foreign debt	

A consistent scenario is not necessarily plausible

We must assess whether these adjustments are realistic. If not, revise the assumptions and run the model again.

Discussion: predict the result

Suppose we increase public investment (IVG), while keeping all other assumptions unchanged.

Recall:

$$IV_t = k_{0,t} GDP_{t-1} + k_{1,t}(GDP_t - GDP_{t-1})$$

$$IV_t = IVP_t + IVG_t$$

Discuss with your neighbour:

1. Does GDP change?
2. Does total investment (IV) change?
3. What happens to private investment (IVP)?

Explain your predictions using the equations. We will check them in the model later.

What do the equations imply?

1. **GDP remains unchanged.**
   We have not changed the sectoral GDP growth assumptions.

2. **Total investment remains unchanged.**
   The investment equation depends on GDP and unchanged coefficients.

3. **Private investment falls by the increase in public investment.**

$$IVP_t = IV_t - IVG_t$$

Why?

In this model, higher public investment changes the composition of investment. It does not automatically increase total investment or GDP.

Julia and the project

Where does each part live?

File or folder	Purpose
`settings.jl`	Base year, horizon and paths
`data/base_year.jl`	Starting values and calibration
`data/assumptions.jl`	Growth rates and parameters
`data/projection_paths.jl`	Inputs for each projection year
`sets.jl`	Sectors and other categories
`Merged_model/variables.jl`	Variable declarations
`Merged_model/modules/`	Economic equations
`run.jl`	Data construction and model run

Julia recap: names and assumptions

```
government_investment_growth = Dict(
    :infrastructure => 0.050,
    :education      => 0.050,
    :health         => 0.050,
)
```

- `:infrastructure` is a symbol used as a dictionary key.
- `0.050` means annual growth of 5 percent.
- Names and capitalisation must match across the files.

Discussion

Which entry would you change to increase education investment growth?

Julia recap: reading an indexed equation

```
IVGI[i = public_investment_types, t = proj],
    IVGI[i, t] ==
        (1 + ivggr[i, t]) * IVGI[i, t - 1]

IVG[t = proj],
    IVG[t] ==
        sum(IVGI[i, t] for i in public_investment_types)
```

- *i* selects a category and *t* selects a year.
- == writes a model equation inside the block.
- `sum` aggregates the components.

Outside the block, = assigns a value. Here it also defines index ranges.

Julia recap: a year-specific assumption

```
:infrastructure => Dict(
    :default => 0.050,
    2027     => 0.150,
),
```

- In a full assumption path, `:default` supplies other years.
- Infrastructure investment grows by 15 percent in 2027.
- It returns to 5 percent growth in subsequent years.

Returning to the original growth rate does not restore the original level.

When Julia reports an error

1. **Read the message.**
   What does Julia say is wrong?

2. **Locate the code.**
   Find the file and line number given in the error code.

3. **Check your latest change.**
   Compare it with a working example.

4. **Fix one thing and run again.**
   Ask for help if you are unsure.

When asking for help

Show the error message and your change. Explain what you were trying to do.

Baseline and scenarios

Running the baseline

Start by running the model with the baseline assumptions.

1. Check the base year and projection horizon in `settings.jl`.
2. Review the assumptions in `data/assumptions.jl`.
3. Open `run.jl` and set `run_scenarios = false`.
4. Save your files. In `run.jl`, press `Ctrl + A`, then `Ctrl + Enter`.

The baseline provides the reference for our later comparisons.

Reading the baseline results

When the calculation finishes, open:

`Merged_model/output/simulation_report.html`

1. Identify the projection years.
2. Locate a variable and check its meaning and units.
3. Read its values across the projection.

The baseline

These results follow from the starting data, the baseline assumptions and the model equations.

From baseline to scenarios

The baseline describes one set of assumptions. Scenarios let us examine alternatives.

Baseline	Scenario
Our reference assumptions and their implied results.	Selected assumptions change. The model calculates new results.

A clear comparison

Keep the baseline unchanged. Start with one changed assumption so its effects are easier to follow.

Defining and running a scenario

1. Open `shocks/scenario.jl`.
2. Give the scenario a name and specify the assumption, component and year to change.
3. Enter the new value. Unspecified assumptions retain their baseline values.
4. Save the file. In `run.jl`, set `run_scenarios = true` and run the file.

Open the comparison report when the calculation finishes:

`shocks/Output/scenario_report.html`

Comparing the results

Compare the same variable in the same year.

- **Levels**
  The value of the variable in the baseline and scenario.
- **Differences from baseline**
  How much the scenario value is above or below the baseline value.
- **Labels and units**
  Check the scenario name, legend and axis labels before reading the result.

Keep the comparisons distinct

A difference from the baseline is not the same as growth from the previous year.

Interpreting the results

Explain the differences using the model relationships.

1. **Start with the changed assumption.**
   Identify the variable directly affected.

2. **Follow the connections.**
   Use the equations to explain changes in other variables.

3. **Include unchanged results.**
   Explain why some variables do not respond.

Economic interpretation

The results show the effects of the change under this model's assumptions and closure.

Worked example: higher infrastructure investment

Increase infrastructure investment growth in 2027 from 5% to 15%. Keep other assumptions unchanged.

In `shocks/scenario.jl`:

```
infrastructure_expansion = (
    government_investment_growth = Dict(
        :infrastructure => Dict(
            2027 => 0.150,
        ),
    ),
),
```

Save the change, run the scenarios and open the comparison report.

Worked example: explaining the results

1. **Public investment increases.**
   Infrastructure investment grows faster.
2. **GDP and total investment remain unchanged.**
   Their determining assumptions and relationships are unchanged.
3. **Private investment falls.**

$$IVP_t = IV_t - IVG_t$$

4. **Government borrowing requirements increase.**
   Additional public investment raises expenditure, while revenue assumptions remain unchanged.

Compare these implications with the results in the report.

Exercises

Exercise 1: Change public investment

Create a scenario with infrastructure investment growth of 10% in 2027.

1. Keep the baseline unchanged and enter the new assumption in `shocks/scenario.jl`.
2. Save your changes, run the model with scenarios enabled and open the comparison report.
3. Compare public investment, private investment and government borrowing with the baseline.
4. Write down one difference and use a model equation to explain it.

When you finish

Change the size of the investment increase and compare the results again.

Exercise 2: Change export growth

Create a scenario with industrial export growth of -3% in 2027.

1. Start from the baseline assumptions. Change only industrial export growth.
2. Before running, write down which results you expect to change.
3. Run the scenario and compare exports, private domestic credit and private foreign debt with the baseline.
4. Explain one result using the model's equations. Note whether it matches your expectation.

When you finish

Extend the lower export growth to another year and compare the results again.

Applications and feedback

Survey: Day 1 follow-up

Please complete the short survey.



▸ Open the survey

Thank you for today

Appendix

Reference: disposable income and prices

Code equation(s): 10, 12–13

$$P_t GDP_t = PD_t(C_t + IV_t) + E_t(XPI_t X_t - MPI_t M_t)$$

$$GDY_t = P_t GDP_t + E_t NFP_t + E_t NTRP_t + INDG_t + (GT_t - TG_t) - E_t INFP_t$$

$$GDS_t = P_t GDP_t + E_t(NFP_t - INFG_t - INFP_t) + E_t(NTRP_t + NTRG_t) - PD_t C_t$$

Here GDY is private disposable income in the implemented model. Its meaning differs from the earlier RMSM notation.

Reference: reserves and interest payments

Code equation(s): 20, 27–29

$$R_t - R_{t-1} = d_t \frac{MPI_t M_t - MPI_{t-1} M_{t-1}}{E_t}$$

$$INDG_t = irdg_t NDDG_{t-1}$$

$$INFG_t = irfg_t NFDG_{t-1}, \qquad INFP_t = irfp_t NFDP_{t-1}$$

These are the reserve and interest rules in the current code. Interest depends on previous-period debt.

Reference: notation used in the code

Code	Meaning
`IV, IVG, IVP`	Total, public and private investment
`C, CG, CP`	Total, government and private consumption
`GDY`	Private disposable income
`P, PD`	GDP deflator and domestic expenditure price index
`TG, GT`	Government revenue and transfers to the private sector
`NDDG`	Government non-bank domestic debt
`DCG, DCP`	Government and private domestic credit
`NFDG, NFDP`	Government and private foreign debt
`E`	Exchange rate, domestic per foreign currency unit

Reference: teaching materials

- First workshop: Lecture 1 and Julia crash course.
- Current model: `Model/Merged_model/modules/`.
- Assumptions: `Model/data/assumptions.jl`.
- Scenarios: `Model/shocks/scenario.jl`.
- Extension guide: `Course/Guides/Guide_Indexed_Extensions.docx`.

github.com/JonathanWPedersen/Vietnam-Merged-Model

Model equations checked against the repository on 8 November 2026.